THE HALTING PROBLEM OF THE MACHINE $1RB1LA_0RC0RD_1LC1LA_1RE1RD_1LF0LA_---_0LE$: A CONDITIONAL NON-HALTING THEOREM AND THE RECORD-TIMETABLE OBSTRUCTION

ABSTRACT. We give a complete structural analysis of the 6-state, 2-symbol Turing machine $M =$ `1RB1LA_0RC0RD_1LC1LA_1RE1RD_1LF0LA_---_0LE` and prove, conditional on a single precisely-stated computational assumption (master closure, Assumption M), that it never halts from the blank tape; unconditionally, non-halting holds over all runs whose round classes stay in the exhaustively verified range ($h \le 2000$, sampled to $1.2 \cdot 10^7$) — a doubly-exponential horizon whose step-count scaling is labeled as empirical (Proposition on the envelope). The analysis proceeds by a macroscopic phase calculus between visits of state D to blank cells, a closed family of recurrent anchors, a halving-chain prologue, and a *round law* proved by strong induction on the 2-adic class $h = \nu_2(\alpha + 1)$ of the leading counter, via a self-similar round decomposition (halving cascade, universal ladder, master segment). Magnitude decisions are handled unconditionally by a dominance inequality (stated as Lemma D of the companion derivation paper); the halting branch is structurally unreachable for $h \ge 82$ and perturbation-robust; $\Delta t(h) \in [2, 9]$ on all verified classes. The single remaining assumption is shown to be equivalent, by a finite-state reduction, to the continuation of a record timetable of the autonomous 2-adic walk, whose 32-bit slice is certified acyclic by a finite experiment (with non-eventual-periodicity beyond the slice stated as a conjecture) — the precise, irreducible core of the problem. Two further groups of results, unconditional with respect to Assumption M, are proved along the way: a mod-3 ledger — from step 54 on, every phase configuration of M has run-sum divisible by 3, so an entire half-space of configurations (including the Conway-chain entry $S(3, 1, 3)$) is unconditionally unreachable and the re-entry and anchor congruences hold as theorems — and a complete birth-sequence theory: a halt equivalence (M halts iff some S-even birth value $A = x + y/2 + 1$ is a power of two; both directions algebraic), sharpened by *root reduction* to "iff some *root* birth is a power of two" (descendant births halve exactly, preserving odd parts); *all E1-reentry roots are killed unconditionally* ($3 \mid A$, a corollary of the ledger; 90.5% of all roots); a *parity kill of dangerous keys* ($q \equiv 1 \bmod 144 \wedge q > 1 \Rightarrow q \ge 145 \Rightarrow$ not a power; 19 of the 20 observed classes permanently safe by key definition); and the *drift law* M$''$ — eight window families with the unified form $A = 3 \cdot 2^{k-1}w + 2^j\mu$ at window positions 13/26/50/99/110/194/219/220, every family parity-killed, with the $m \le 40$ synthetic assertion table (60 segments, 0 powers, 56/56 exact F_{26} identities) — which compresses the residual of Assumption M into a single arithmetic sentence in birth coordinates: the conditional chain of the main theorem is thereby refined to "never halts $\Leftarrow$ M$''$ + class preservation", with the three precise gaps to an unconditional proof (window-enumeration completeness = M; conditionality of class preservation; $m > 25$ as a constructed-anchor property) stated explicitly. Per-depth raw-reproduced witnesses (through depth 3) show that birth guards fail at every finite depth, stated at witnessed level. All other components are proved algebraically or verified exhaustively over declared finite domains; every claim is classified in a ledger (algebraic / exhaustive / assumption-dependent).

1. INTRODUCTION

Let M denote the Turing machine with states $A, \ldots, F$ and transition table of Table 1, in the standard notation where, e.g., `1RB` means: write 1, move right, go to state B, and where the single undefined transition (state F reading 0) is the halting transition. The machine is one

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of the remaining undecided instances of the 6-state Busy Beaver problem; to our knowledge no analysis of it has been published.

	read 0	read 1
A	1RB	1LA
B	0RC	0RD
C	1LC	1LA
D	1RE	1RD
E	1LF	0LA
F	HALT	0LE

TABLE 1. The transition table of M.

Theorem 1.1 (Main, conditional form)**.** *M, started on the blank tape, never halts, conditional on Assumption 5.10 (master closure for all classes).*

A second form of the conditional chain — "never halts ⇐ drift law M″ + class preservation" — is established in §4 (Corollary 4.16); by Proposition 4.14(d) the two forms carry the same non-halting content within the established architecture.

Proposition 1.2 (unconditional envelope, verified range; empirical status made explicit)**.** *Unconditionally, M does not halt within any run whose round classes stay in the exhaustively verified range (all classes $h \le 2000$; sampled agreement through $1.2 \cdot 10^7$) — in step-count terms, non-halting is proven over a horizon that is doubly exponential in the class index (below $2^{2^{2000}}$ in the record-climb heuristic).*

Status of the envelope. The finite part is unconditional: over the verified range the round table is single-path and carries no halting block (Theorem 5.3, computational components of §9); the anchors climb by $n' \ge n + 6$ per exit, and the record timetable of the residual climbs by a measured factor ≈ 4 per class level, so the first arrival of class h requires a number of rounds growing like 4^h, each of supercubic cost in the anchor index — hence the doubly-exponential horizon. The record climb, however, is a *measured* law of the autonomous walk (Theorem 9.1), not an unconditional theorem; we therefore state the envelope as a proposition whose step-count translation carries this explicitly labeled empirical component. A paper proof of an explicit lower bound $T(h) \ge 2^{c \cdot 2^h}$ (assembling $A_j \le \tau_j^2$ with the band $\Delta\alpha \le 2h+6$) is left open. Independently of the class range, the shape of the unconditional envelope is sharpened by Theorems 4.3 and 4.6: every configuration with $\Sigma \not\equiv 0 \pmod 3$ is unreachable from step 54 on (an entire half-space, with no appeal to any verified class range), and "M never halts" is equivalent to "no power-of-two S-even birth ever occurs" — the verified domain of the latter equivalence being the first 10^7 phase events (2,908 births, none a power of two; grade B); by root reduction (Theorem 4.8) it is further equivalent to "no power-of-two *root* birth ever occurs", whose verified domain is: 30,791,021 root births over 10^7 anchors with 0 powers, all 20 observed classes safe (18 by key definition + Lk|j3 single-point + L3|j4 family form), all eight dangerous window families killed through constructed anchors $a \le 300$, and the $m \le 40$ assertion table all passing (grade B). □

The proof occupies Sections 2–8. Its skeleton:

(1) (§2) A phase calculus for the instants when the head, in state D, reads a 0. Between two such instants the machine executes a fully determined finite block of elementary steps, and the possible tape shapes at these instants form a finite grammar with ten transition rules (nine non-halting event edges and the halting block; Lemma 2.3). The unique halting path passes through a single grammar configuration $G(3, \cdot)$ (Lemma 2.4).

(2) (§3) The blank-tape run reaches at step 54 an *anchor* $P(3)$, and every anchor $P(n)$ is carried, by an explicit halving chain of length $m = \nu_2(n+3)$, into a standard family $\Phi(w, m)$, $n + 3 = 2^m w$, w odd (Theorem 3.4).
(3) (§4) Unconditionally with respect to Assumption 5.10: the mod-3 ledger (an entire half-space of configurations unreachable from step 54 on), the halt equivalence via power-of-two births and its root-reduction sharpening, the E1-reentry kill, the parity kill of dangerous keys, the L-exit class snapshot, and the drift law M″ — yielding the second form of the conditional chain, "never halts ⇐ M″ + class preservation" (Corollary 4.16), with the three precise gaps stated (Remark 4.17).
(4) (§5–5.2) The Φ-trajectory visits *landmarks* $L(\alpha, t; s)$ indexed by a leading counter $\alpha \equiv 1 \pmod 6$ and a time parameter t; the evolution of (α, t) depends only on the class $h = \nu_2(\alpha + 1)$ (Theorem 5.3, the *round law*), proved by strong induction on h via a halving cascade (Lemma 5.4), a universal ladder (Lemma 5.8), and a uniform master segment (Assumption 5.10, uniformity clause — Remark 5.11).
(5) (§6–8) Exhausted-tail configurations exit to the next anchor with $n' \geq n + 6$ and $n' \equiv 3 \pmod 6$ (Theorem 6.1), and no reachable configuration is the halting one (Theorem 7.1). The anchors therefore climb forever.

Throughout, tape contents are described by *runs* (maximal blocks of 1s) separated by *gaps* (blocks of 0s); all configurations live on the right-infinite half of the tape beyond a fixed left sentinel, which the machine never crosses (this follows from the left-moving transitions returning before the sentinel; it is confirmed by every invariant below).

2. The phase calculus

2.1. **Events and configurations.**

Definition 2.1 (event)**.** An *event* is an instant at which the head, in state D, reads a 0. Let $\mathcal{C}$ denote the set of tape shapes (with head position) that can hold at an event.

Definition 2.2 (grammar)**.** Write a configuration as a list of runs $(r_1, r_2, \ldots, r_k)$, $k \geq 2$, with specified gaps between them and a marked head cell (the 0 being read). The following four types exhaust the configurations that arise on the orbits considered:

- $R(a, b)$: runs (a, b), gaps $[1]$, head on the fresh 0 *immediately right* of the right run;
- $S_k(x, y, z, \ldots)$: $k \geq 3$ runs, gaps $[1, 2, 1, 2, \ldots]$, head on the *left* 0 of the 2-gap between runs 2 and 3;
- $G_k(a, b, \ldots)$: $k \geq 2$ runs, gaps $[3, 1, 1, \ldots]$, head on the *middle* 0 of the 3-gap between runs 1 and 2;
- $L_k(r_1, r_2, r_3, \ldots)$: $k \geq 3$ runs, gaps $[1, 1, 1, \ldots]$, head on the single-0 gap between runs 2 and 3.

(Thus L_2 is identified with R: both have the head just right of the block.)

Lemma 2.3 (phase rules)**.** *Between consecutive events, M executes exactly the following transitions, with the stated exact step counts:*

(R)	$R(a,b) \to S_3(a{+}1,\ b{-}2,\ 1)$	$2b+3, \quad b \geq 3;$
(S≥)	$S_k(\dots,x,y,z,\dots) \to S_k(\dots,x{+}1,\ y{-}2,\ z{+}1,\dots)$	$2y+3, \quad y \geq 3;$
(S2)	$S_k(\dots,x,2,z,\dots) \to G_{k-1}(\dots,x{+}1,\ z{+}1,\dots)$	$7;$
(S1)	$S_k(\dots,x,1,z,\dots) \to L_{k-1}(\dots,1,\ x{+}z{+}3,\dots)$	$2x+z+14;$
(G≥)	$G_k(a,b,\dots) \to S_{k+2}(1,\ a{-}3,\ 1,\ b{+}1,\dots)$	$2a+3, \quad a \geq 4;$
(G3)	$G_k(3,b,\dots) \to G_{k+1}(1,\ 1,\ b{+}1,\dots)$	$9;$
(G2)	$G_k(2,b,\dots) \to L_{k+1}(1,\ 3,\ b{+}1,\dots)$	$16;$
(G1)	$G_k(1,b,\dots) \to \mathsf{HALT}$	$6;$
(L≥)	$L_k(r_1,r_2,r_3,\dots) \to L_k(r_1{+}1,\ r_2,\ r_3{-}1,\dots)$	$2r_2+5, \quad r_3 \geq 2;$
(L1)	$L_k(r_1,r_2,1,\dots) \to R/S_{k-1}(r_1{+}1,\ r_2,\dots)$	$2r_2+5.$

In (S1) the result is R when $k=3$ and L_{k-1} when $k \geq 4$; in (L1) it is R when $k=3$ and S_{k-1} when $k \geq 4$. The halting block (G1) ends with F reading 0; the block (G3) is an ordinary event block whose target $G(1,\cdot)$ halts in the next block.

Proof sketch. Each rule is obtained by executing the transition table from the stated configuration until the next event. We illustrate rule (R) for $R(3,3)$ (tape $\dots 1^3\,0\,1^3\,0^\infty$, head on the fresh 0, state D; the trace below is a step-level recording of the machine):

t	state/head	effect
0	D@7	tentative write 1, right, E
1	E@8	write 1, left, F
2	F@7	*erase* (`0LE`), `left`, E
3	E@6	*erase* (`0LA`), `left`, A
4, 5	A@5, A@4	march left through the right run
6	A@3	read the gap 0: *fill* it (`1RB`), `right`, B
7	B@4	write 0 (`0RD`), `right`, D
8	D@5	march right writing 1s
9	D@6	next event: $S_3(4,1,1)$

The net effect is a five-phase mechanism: (i) D's tentative extension of the right run is *undone* by the E–F–E–A recoil; (ii) A marches left across the right run; (iii) A fills the inter-run gap with 1, growing the first run to length $a{+}1$; (iv) B zeroes one cell of the old right run, creating the new 2-gap; (v) D marches right, rebuilding, so that the old right run of length b leaves a residue $b-2$ to the left of the new gap and a fresh run of length 1 to its right: $S_3(a+1,b-2,1)$, at total cost $2b+3$. The remaining rules follow by the same kind of shuttling; each was derived stepwise from Table 1. Rule-level faithfulness is machine-checked in two complementary ways: (i) the rule checker `rulecheck.mjs` compares every rule against the raw machine on parameter grids ($a \in \{1,\dots,20\}$, $b \in \{2,3,5,10\}$, tails of length 0–2): 564/564 rows agree exactly (this includes the boundary cases $y \in \{1,2\}$, $a \in \{1,2,3\}$, and the halting case (G1)); (ii) trajectory-level comparison of the phase engine against the raw oracle over 956 anchor trajectories ($n = 2..957$): 955 agree field-by-field, including 16 halting anchors with exact halt times; the single exception is $R(2,2)$, whose successor ($b = 2 < 3$) lies outside the declared grammar and is flagged, not mis-computed (direct simulation halts at +22). An earlier version of the table carried a folded (G3)→ HALT row and an incorrect (G2) image $L(1,b{+}1,b{+}1)$; against that version the checker reported 531 ok / 21 bad, the bad rows being precisely (G2)/(G3) — the correction here follows the raw-machine side of that checker. □

Lemma 2.4 (halting criterion)**.** *M halts from an event configuration iff the phase trajectory reaches $G_k(1, b, \dots)$ for some b and tail; equivalently, iff it reaches a G-state with leading head $a \in \{1, 3\}$ (rule (G3) carries $G(3, \cdot)$ to $G(1, \cdot)$ in one block). On every orbit considered below, the leading heads of all G-states are ≥ 2 and $\neq 3$, so no other rule can produce a halt.*

Proof. The only undefined transition of Table 1 is F reading 0. An exhaustive check of the ten rules shows the state F reads a 0 in the interior of an event block in exactly one grammar situation, namely the configuration $G(1, b, \dots)$ (rule (G1)); in all other blocks F reads 1s or the block ends at the next event first. Rule (G3) is the sole in-rule source of $G(1, \cdot)$. □

2.2. **Sweeps.** For efficiency we also use *swept* versions of the rules: consecutive applications of (S≥) with the same x-growth collapse into one step with closed-form cost. If y is odd, $j = (y-1)/2$ applications carry $(x, y, z) \mapsto (x + j,\, 1,\, z + j)$ at total cost $\sum_{i=0}^{j-1}(2(y - 2i) + 3)$; if y is even, $j = (y-2)/2$ applications carry $(x, y, z) \mapsto (x + j,\, 2,\, z + j)$. These identities are elementary and are used silently below.

3. Anchors and the prologue

Definition 3.1 (anchor)**.** For $n \geq 2$ let $P(n)$ denote the configuration $0^\infty 1^n 0 1^n 0^\infty$ with the head in state D on the fresh 0 immediately right of the right run; in grammar terms $P(n) = R(n, n)$.

Lemma 3.2 (entry)**.** *Started on the blank tape, M reaches $P(3)$ at step 54.*

Proof. Direct simulation of the 54 steps: the machine builds the pattern $1^3 0 1^3$ by the *A-B-C-D-E-F* loop and lands in state D on the fresh 0. (Step-level trace in the repository; verified by two independent simulators.) □

Definition 3.3 (the family Φ)**.** For odd $w \geq 3$ and $m \geq 1$:

$$\Phi(w, m) \;=\; G\big(w - 1,\; w - 1,\; 2w,\; 4w,\; \dots,\; 2^{m-1}w,\; 2^m w - 1\big).$$

Theorem 3.4 (prologue)**.** *Let $n \equiv 3 \pmod 6$, $n \geq 3$ (this holds for every anchor on the orbit of Theorem 1.1; in particular n is odd), and write $n + 3 = 2^m w$ with w odd, $m \geq 1$. Then the phase trajectory from $P(n)$ reaches $\Phi(w, m)$; explicitly,*

$$R(n, n) \;\to\; R(1,\, 2n + 2) \;\to\; G(n + 2,\, n + 1) \;\to^{m \text{ halvings}}\; \Phi(w, m),$$

where the halving steps act on the two leading runs while the tail is carried through, and the leading heads follow the closed form

$$a_i \;=\; 2^{m-i}\, w - 1, \qquad i = 0, 1, \dots, m. \tag{1}$$

Proof. Since n is odd, rule (R) gives $R(n, n) \to S_3(n + 1,\, n - 2,\, 1)$ with middle run $y = n - 2$ *odd*; the sweep drives y down to 1 in $j = (n - 3)/2$ steps, and rule (S1) yields $R(1,\, x + z + 3)$ with $x = n + 1 + j$, $z = 1 + j$, i.e.

$$R(1,\, 2n + 2).$$

Rule (R) on $b = 2n + 2$ gives $S_3(2,\, 2n,\, 1)$ with $y = 2n$ *even*; the sweep drives y down to 2 in $n - 1$ steps, and rule (S2) yields

$$G\big(2 + (n - 1) + 1,\; 1 + (n - 1) + 1\big) \;=\; G(n + 2,\, n + 1).$$

Now let $G(a, b, \mathcal{R})$ be a halving state of the chain started at $G(n+2, n+1)$, i.e. $a+1 = 2^{m-i}w$ with $0 \leq i \leq m$. Since w is odd, a is odd exactly when $i < m$, so the halving map (rules (G≥) followed by the L/S shuttles of §2, whose composite on the two leading runs is $(a, b) \mapsto \big((a-1)/2,\, (a-1)/2\big)$ with b entering the accumulating geometric tail) applies for $i = 0, \dots, m-1$.

Starting from $a_0 + 1 = n + 3 = 2^m w$ the heads satisfy (1), terminating at $a_m = w - 1$ with accumulated tail $(2w, 4w, \ldots, 2^{m-1}w)$ and terminator $2^m w - 1$: exactly $\Phi(w, m)$. □

Remark 3.5. Theorem 3.4 is stated for $n \equiv 3 \pmod 6$, the congruence class of every anchor reached from $P(3)$ (Theorem 6.1, §8); for even n the initial sweep branches differently and the chain above does not apply. The theorem is algebraic and holds for all $m \geq 1$; it was additionally machine-witnessed for all $n \equiv 3 \pmod 6$, $n \leq 6003$ (1001 values), and for $m \in \{13, \ldots, 1200\}$, $w \in \{3, 9, 21\}$.

4. The mod-3 ledger and the birth sequence

This section contributes a chain of results that are *unconditional* with respect to Assumption 5.10: a mod-3 ledger for the whole orbit (Theorem 4.3), a halt equivalence in terms of power-of-two births (Theorem 4.6) with its root-reduction sharpening (Theorem 4.8), unconditional safety of all E1-reentry root births (Lemma 4.9), a per-key parity kill (Theorem 4.12), the L-exit class-table snapshot (Proposition 4.11), the m-record timetable (Proposition 4.13), the drift law M″ — the final explicit form of the residual (Proposition 4.14) — and the synthetic assertion table through $m = 40$ (Proposition 4.15); witnessed-level propositions on the pollution of birth guards follow (Propositions 4.18 and 4.19). Together they seal the mod-3 side unconditionally, kill all E1-reentry roots, kill every identified L-exit family by pure parity, and condense the entire residual content of halt-exclusion into the single statement M″, whose precise three-ring gap to an unconditional proof is spelled out (Remark 4.17).

Definition 4.1 (ledger invariant I_0)**.** For a configuration C let $\Sigma(C)$ denote the sum of its runs. I_0 is the set of configurations satisfying: (i) $\Sigma \equiv 0 \pmod 3$; (ii) domain: $R(a, b)$ with $b \geq 3$; $S(x, y, z, \overline{r})$ with $y \geq 1$; $G(a, b, \overline{r})$ with $a \geq 2$; $L(r_1, r_2, r_3, \overline{r})$ with $r_2 \geq 3$ and $r_3 \geq 1$; (iii) boundary guards BG1: $G(2, b, \overline{r})$ requires $b \geq 2$ and $b \equiv 2 \pmod 3$; BG2: $S(1, 2, z, \overline{r})$ requires $z \equiv 1 \pmod 3$. Both guards are *forced* by closure: BG2 because (S2) births $G(2, z{+}1)$, which must pass BG1; BG1 because the Σ-increment of the (G2) edge in the derivation engine's form is $b{+}1$ (at the raw level the increment is constantly $+3$; Remark 4.4). Under $\Sigma \wedge$BG2 the three-run class $S(1, 2, z)$ is empty ($z \equiv 0$ contradicts $z \equiv 1$); the class $S(1, 2, z, \overline{r})$ with $k \geq 4$ is nonempty by tail freedom, and the orbit runs precisely there ((G$\geq$) at $a = 5$ births $S(1, 2, 1, b{+}1, \overline{r})$, $z = 1$).

Lemma 4.2 (unique reachable value of $G(2, \cdot)$)**.** *The grammar-reachable configurations $G(2, b, \overline{r})$ are exactly those with $b = 2$ (including the geometric-tail family $\Phi(3, m)$).*

Proof. The only in-rule source of a G-state is (S2), $S(x, 2, z, \overline{r}) \to G(x{+}1, z{+}1, \overline{r})$; head 2 forces $x = 1$ at the sweep terminal. An S-entry with $x = 1$ can only come from (G$\geq$) ((R) births $x = a{+}1 \geq 2$; (L1) births $x = r_1{+}1 \geq 2$; (S$\geq$) sweeps increase x), i.e. $S(1, a{-}3, 1, b{+}1, \overline{r})$ with $z = 1$ fixed; reaching $y = 2$ with x still 1 forces the sweep length $j = 0$, i.e. $a = 5$, yielding $S(1, 2, 1, b{+}1) \to G(2, 2, b{+}1, \overline{r})$. □

Theorem 4.3 (mod-3 ledger; unconditional w.r.t. Assumption M)**.** *(a)* Closure. *Every member of I_0 steps to a member of I_0 under each of the ten rules of Lemma 2.3, the only exceptions being the halt forms $G(3, b, \overline{r})$ (classified, not violations). (b)* Seed. *The blank-tape orbit enters $R(3, 3) \in I_0$ at step 54 ($\Sigma = 6$). (c)* Conclusion. *Every phase (grammar) configuration of M from step 54 on satisfies $\Sigma \equiv 0 \pmod 3$. Consequently every re-entry $R(1, b)$ has $b \equiv 2 \pmod 3$; every anchor $R(n, n)$ has $n \equiv 0 \pmod 3$; any halt must occur at a $\Sigma \equiv 0$ form $G(3, b, \overline{r})$; and every configuration with $\Sigma \not\equiv 0$ — including the Conway-chain entry $S(3, 1, 3)$ ($\Sigma = 7$) — is unreachable, an entire half-space. I_0 does not exclude halting (the halt forms $G(3, \cdot)$ can lie in I_0); the entire residual content of halt-exclusion lies on the 2-adic birth side (Theorem 4.6).*

[1]

Proof status (grade A$^-$, closure part): the closure is given by per-edge elementary affine identities plus guard transfers — (R): $(a{+}1)+(b{-}2)+1=a+b$; (S$\geq$) and (L$\geq$): translations; (S2): $(x{+}1)+(z{+}1)=x{+}z{+}2$; (S1), both branches: $+3$; (G$\geq$): $1+(a{-}3)+1+(b{+}1)=a+b$; (G3): $\Delta=0$; (G2): Remark 4.4; (L1), both branches: $+1$ — each hand-checkable, and re-verified by machine sweeps: the author's boundary enumeration of 219,398 I_0 member-steps plus 2^{40}-scale random member-steps (the assertion total fluctuates by about ± 60 with the random sampling; the deterministic core reproduces digit-for-digit), and an independent heterogeneous closure grid of 20,508 member-steps with run counts up to $k \leq 10$ and $a=0$ probes. The reduction of closure to bounded run count is carried by locality (each rule rewrites only the first three runs; both guards read only the first three). Event-level raw fidelity was independently re-verified (all ten rules, boundary grids plus 2^{24}-scale random); the seed is triple-sourced (two simulators plus an independent raw-machine parse). The orbit side is a redundant confirmation (the induction already yields the all-times statement): 10^7 phase events with zero violations (406 anchors, all $n \equiv 0$; 1,485 re-entries, all $b \equiv 2$); an independent re-check of 5×10^5 events with zero violations (91 anchors, 332 re-entries); the trajectory library of 1,357,014 states with zero violations (R-family decomposition $825 = 547 + 150 + 128$, all $a+b \equiv 0$). The 112 halt forms $G(3,\cdot)$ mentioned above are an enumeration count of the closure grid, not an orbit observation (the orbit count of $G(3,\cdot)$ is 0).

Remark 4.4 (raw fidelity domain of (G2))**.** Raw truth: the (G2) edge maps $G(2,b,\overline{r}) \to L(1,3,b{+}1,\overline{r})$ for *all* $b \geq 2$ (raw-verified for $b \in \{2,3,5,8,11,14\}$ with three tail shapes; verbatim the (G2) row of Lemma 2.3, whose rule checker 564/564 also covers (G2) at each b), with Σ-increment constantly $+3$ — so the mod-3 ledger is preserved by the true machine on all of I_0 *without appeal to BG1* (under the raw identity BG1 becomes superfluous; only the engine-level bookkeeping needs it). The derivation engine of the ledger analysis suite carries the simplified image $L(1,b{+}1,b{+}1,\overline{r})$, which agrees with the raw machine precisely at $b=2$; by Lemma 4.2 this is the unique grammar-reachable value, so the distortion site $G(2,b\neq 2)$ is unreachable, and even if it were entered (started from an off-grammar configuration) the increment $+3$ would still preserve the ledger. Accordingly, the reading "ten rules = raw machine" must be qualified: the rule table is event-level raw-faithful on the grammar domain; on the off-grammar layer ($G(2,b\neq 2)$) the engine uses the simplified image, with no effect on the ledger.

Definition 4.5 (S-even birth)**.** An event whose configuration is an $S(x,y,z,\overline{r})$ with y even ($y \geq 2$) is called an *S-even birth* (the entry of that even sweep); its *birth value* is

$$A \;=\; x + \tfrac{y}{2} + 1.$$

Theorem 4.6 (halt equivalence via power-of-two births)**.** *M (started on the blank tape) halts if and only if some S-even birth value $A = x + y/2 + 1$ on its phase trajectory is a power of two; equivalently $A \in \{4,8,16,\dots\}$ (always $A \geq 3$, since $x \geq 1$ and $y \geq 2$; for $A=3$ the sweep terminal has head 2 and feeds the non-halting (G2) channel).*

Proof. First, the *sweep invariant*: (S$\geq$) preserves A, since $(x{+}1)+\frac{y-2}{2}+1 = x+\frac{y}{2}+1$; a birth sweep terminates at $S(x',2,z')$ with $x' = x+\frac{y-2}{2}$, and (S2) lands in the G-state with head $x'{+}1 = A-1$. ($\Leftarrow$) Let some birth have $A = 2^t$ ($t \geq 2$). Its sweep terminal G-state has head 2^t-1 (odd, ≥ 3). For $t=2$ this is already the halt form $G(3,\cdot)$. For $t \geq 3$ the

[1] Granularity of the halt segment: the folded block (G3)+(G1) (15 steps total) contains, at $+9$ steps, the intermediate event $G(1,1,b{+}1,\overline{r})$ — the target of (G3), outside the G-domain ($a \geq 2$) of I_0 but still with $\Sigma \equiv 0$ (the (G3) Σ-identity has $\Delta = 0$; spot-verified against the raw machine). The universal statement of Theorem 4.3(c) is thus pinned to *phase (grammar) configurations*.

head is ≥ 7 and odd, and the halving composite of Theorem 3.4 (on the two leading runs, $(a,b) \mapsto \left(\frac{a-1}{2}, \frac{a-1}{2}\right)$ with the tail carried verbatim) applies at each step: the head chain runs $2^t - 1 \to 2^{t-1} - 1 \to \cdots \to 7 \to 3$ (every head on the chain has the form $2^s - 1 \geq 3$, never the value 5, never the $G(2,\cdot)$ channel), ending at the halt form $G(3,\cdot)$; by Lemma 2.4, M halts. ($\Rightarrow$) Suppose M halts. By Lemma 2.4 the trajectory reaches a G-state with head 3; the only in-rule source of $G(3, z{+}1, \overline{r})$ is (S2) acting on $S(2, 2, z, \overline{r})$. That S-state has $y = 2$ and lies inside some even sweep (or is itself its entry); the sweep invariant gives the birth value at that sweep's entry $A = 2 + \frac{2}{2} + 1 = 4 = 2^2$, a power of two. □

Proof status (grade A$^-$, both directions): both directions are elementary algebra, resting on rule faithfulness and the uniqueness of the halt form (heads $\in \{1, 3\}$ with only the (G3)/(G1) exits, raw-verified broadly); rule faithfulness is the B-grade check (rulecheck 564/564, 956 anchors) plus the independent event-level raw re-verification. Empirical companion (grade B): in the first 10^7 phase events there are 2,908 S-even births, none a power of two; an independent re-check of 5×10^5 events finds 647 births, none. The entire content of Assumption M condenses to the single sentence "no power-of-two birth ever occurs". The next two subsections compress that sentence further, onto *root* births and then onto the dangerous keys of L-exits, ending with the explicit form M$''$ of the residual.

4.1. **Root reduction and E1-reentry safety.**

Definition 4.7 (root and descendant births)**.** A birth (Definition 4.5) is exactly one of the following by its entry edge: an *E1 root*, entering the even sweep via (R) from $R(a,b)$ with b even, $A = a{+}b/2{+}1$ — on the orbit, the re-entries $R(1,b)$ (b even) and the anchor initializations $R(n,n)$ (n even); an *L-exit root*, entering via (L1) from $L(r_1, r_2, \overline{r})$ with r_2 even, $A = \sigma + r_2/2 + 1$ with $\sigma = r_1{+}r_3$ (L_3 relays through $R(\sigma, r_2)$ with the same formula: $L_3 \to R(\sigma, r_2) \to S(\sigma{+}1, r_2{-}2, 1)$, $A = \sigma + 1 + \frac{r_2-2}{2} + 1 = \sigma + r_2/2 + 1$); or a *descendant*, entering via (G$\geq$) from $G(a, \overline{r})$ with a odd ≥ 5, of value $A' = (a{+}1)/2$ — where $a{+}1$ is exactly the birth value that produced that G-state (the (S2) landing head is $A - 1$; sweep invariant in the proof of Theorem 4.6).

Theorem 4.8 (root reduction)**.** *Every descendant birth has exactly half the value of its chain's root birth; inductively,* the odd part of every birth equals the odd part of its chain's root birth. *In particular,*

$$M \text{ halts} \iff \text{some root birth has } A = 2^t \ (t \geq 2).$$

Proof. By Definition 4.7 descendant values $A' = (a{+}1)/2 = A/2$ halve exactly, so odd parts are chain-invariant. ($\Leftarrow$) Let a root birth have $A = 2^t$: its even sweep lands via (S2) in the G-state with head $2^t - 1$, after which the halving composite (Lemma applied as in Theorem 3.4) runs $2^t - 1 \to 2^{t-1} - 1 \to \cdots \to 7 \to 3$, reaching the halt form $G(3,\cdot)$ after $t - 2$ steps — every head on the chain has the form $2^s - 1 \geq 3$, independently of b and the tail (carried verbatim); by Lemma 2.4 M halts. ($\Rightarrow$) This is the ($\Rightarrow$) of Theorem 4.6: a halt gives some S-even birth equal to 2^t; if that birth is a descendant its odd part is 1, hence its chain root has odd part 1, i.e. the root is a power 2^s. □

Proof status (grade A$^-$). The classification algebra is as above (resting on rule faithfulness and the halving composite); the exhaustive machine evidence for the provenance classification (grade B, adversarial re-audit ref-b1): over 2×10^7 uncollapsed events / 572 anchors, all 4,120 births originate from R/L/G only (8,576,839 S-interior repeats correctly deduplicated), every R tagged L_3 comes from the $L_3 \to R$ relay; every G-entry head $a{+}1$ equals the S-entry birth value that produced it (0 violations); descendant inheritance $A' = (a_{\text{prev}}{+}1)/2$ (0 violations); descendant odd part = nearest root odd part (0 violations); the uncollapsed and collapsed engines agree

term-by-term on the (ty, A) root sequences (0 mismatches; births10's 600-anchor cross-check independently reproduced); an independent raw-machine simulator reproduces the first 12 anchor transitions of the blank-tape run exactly in both n' and cost. □

Lemma 4.9 (E1-reentry root safety)**.** *Every re-entry $R(1, b)$ on the orbit satisfies $b \equiv 2 \pmod 3$ (Theorem 4.3, unconditional). If moreover b is even (the only case that births), then $b \equiv 2 \pmod 6$, $b \geq 8$, and*

$$A = \tfrac{b}{2} + 2 \;\equiv\; 1 + 2 \equiv 0 \pmod 3, \qquad A \geq 6.$$

Hence no E1-reentry root birth is ever a power of two — *unconditionally with respect to Assumption M.*

Proof. $b \equiv 2 \pmod 6$ gives $b/2 \equiv 1 \pmod 3$; the domain $b \geq 3$ with $b \equiv 2 \pmod 6$ forces $b \geq 8$, so $A \geq 6$; and $2^t \not\equiv 0 \pmod 3$. □

Empirical companion (grade B, independent counter ref-b2): over 10^6 anchors, all 2,787,614 E1-reentry root births satisfy $3 \mid A$ (0 exceptions; all with $b \equiv 2 \pmod 6$; least odd part 3); all 3,669,365 re-entries (either parity of b) have $b \equiv 2 \pmod 3$.

Lemma 4.10 (sharpened AP power criterion)**.** *Let $D = 2^s u$ (u odd) > 0 and $A_0 = 2^j q_0$ (q_0 odd) with $j < s$. Then the arithmetic progression $\{A_0 + Dk : k \geq 0\}$ contains a power of two* iff *$q_0 = 1$ (i.e. A_0 itself is a power). For $j \geq s$ containment is decidable: $\exists\, t \geq s$ with $2^t \geq A_0$ and $2^t \equiv A_0 \pmod D$ (a discrete logarithm modulo u; a correct implementation must enumerate the full preperiod-s plus period $\mathrm{ord}_u(2)$ structure of $2^t \bmod D$).*

Proof. For $j < s$ every member has $\nu_2(\cdot)$ exactly j and odd part $q_0 + 2^{s-j}uk$, odd and affinely increasing $\geq q_0$; a power (odd part 1) occurs iff $q_0 = 1$. □

4.2. **L-exit roots: the class snapshot, the parity kill, and pairing.**

Proposition 4.11 (L-exit class-table snapshot; declared domain 10^7 anchors)**.** *Within anchor indices $\leq 10^7$ (final value $n \leq 123{,}164{,}085 \approx 1.23\times10^8$, $m = \nu_2(n+3) \leq 25$; collapsed-engine per-event walks, cross-checked term-by-term against the uncollapsed step semantics on 600 anchors, 0 mismatches): **(a)** there are 30,791,021 root births, **none a power of two**; composition (exact counts at 10^6 anchors, independently reproduced digit-for-digit): E1-reentries 2,787,614 (90.53% of roots, killed unconditionally by Lemma 4.9; at the 10^7 scale E1r is covered by the lemma and the L-exits by the direct scan[2]), L-exits 291,483 (9.47%; Lk 145,614 + L3 145,869). **(b)** Every L-exit root falls into one of **20 classes** keyed by the 4-segment key $(\mathrm{ty}, \nu_2(A), A \bmod 144, q \bmod 144)$ (= 17 3-segment coarse keys; the main chain L3\|j2 splits into 4 q-subclasses); each class fits an arithmetic progression $\{A_0 + Dk\}$ (D = gcd of observed differences; fit consistent with all observed members, independently re-verified), with observed modulus family $D = 2^s u$, $u \in \{9, 45\}$, $s \in \{4, 7, 11, 12, 13, 16\}$. Representative classes ($A_0 + Dk$): L3\|j2 $= 76 + 144k$; Lk\|j0 $= 77 + 144k$; Lk\|j3 $= 584 + 1152k$; Lk\|j1 $= 9230 + 18432k$; L3\|j0 $= 583 + 1152k$; L3\|j4 $= 36880 + 73728k$; Lk\|j2 $= 294932 + 589824k$; odd classes $A \equiv \{7, 13, 19, 25, 29, 77, 125\} \pmod{144}$; late classes $A \equiv \{26, 28, 29\} \pmod{144}$ (first at anchors 1,021,640 / 2,043,278 / 4,086,554). **(c) Pairing law.** Every Lk class (σ, r_2) pairs with an L3 class $(\sigma - 1, r_2)$ at the same r_2, with $A_{\mathrm{Lk}} = A_{\mathrm{L3}} + 1$ (14,560/14,560 over all 10^5 anchors, zero violations; the converse fails for 7,305/14,587 L3 roots, which have no Lk partner). **(d)** The table saturates after anchor 4,086,554 (the m=24 record): 5.9×10^6 further anchors produce no new class. Cumulative raw TM steps $\gtrsim 10^{23}$ (re-estimated $\sim 10^{24}$ from the verified 16–$22n^2$ cost band).*

[2]The 10^7 sweep (births7) performs the power check only on L-exit roots (the E1r branch carries no check); the E1r-side evidence chain is Lemma 4.9 (unconditional) plus the full 10^6-anchor direct scan, $3 \mid A$ in all cases.

Certification layers (with the four corrections of the adversarial audit birth-referee incorporated):

- **18 of the 20 classes are unconditionally safe by key definition**: their residue $A \bmod 144$ lies outside $\{2^t \bmod 144\} = \{1, 2, 4, 8, 16, 32, 64, 128, 112, 80\}$, so *no* member — present or future — is a power of two.
- **Lk|j3 ($A \equiv 8 \bmod 144$) is a single-point collision**: its only power, 8, is below the least observed member 584; and $A = 8$ would require $q \equiv 1$, i.e. the *unobserved* key (Lk,3,8,1), not this class.
- **L3|j4 ($A \equiv 16 \bmod 144$) is the unique effective collision class** (its power family $t \in \{4, 10, 16, \dots\}$ is unbounded): its members form the family $A = 9 \cdot 2^{12}W + 16$ (W odd ≥ 1, $r_2 - \sigma = 30$) with odd part $2304W + 1 > 1$ — power-free *conditional on the family form* (25/25 orbital members at 10^5 anchors + 6/6 synthetic members verified individually; direct scan $W \leq 2 \times 10^6$, 0 powers).
- **Artifact reality**: the offline certification artifact (births11) contains **14** (3-segment-key) classes — its member sources omit the three late classes; those three are covered by mod-144 residue-class supersets (independent exhaustive brute-force decider). **AP extrapolation direction:** when D_{fit} is a multiple of the true period, the fitted progression is a *subset* of the class completion, not a superset — a power-free fit does not certify a power-free class (counterexample: the true class $\{80 + 48k\}$ contains 128, while the observations 80, 176, 272 fit the power-free $\{80 + 96k\}$). The actual load carried by the AP fit is consistency with all observed members (verified) and the extrapolation of the L3|j4 family form (explicitly conditional). [3]

Theorem 4.12 (parity kill of dangerous keys)**.** *Let $A = 2^j q$ (q odd) be any birth value; A is a power of two iff $q = 1$. Hence: (i) $q \not\equiv 1 \pmod{144} \Rightarrow q \neq 1 \Rightarrow$ not a power; (ii) $q \equiv 1 \pmod{144} \wedge q > 1 \Rightarrow q \geq 145 \Rightarrow$ not a power; (iii) $j = 0$ (A odd) $\Rightarrow$ not a power. Counting with the full 4-segment keys,* ***19 of the 20 observed classes*** *have q-segment $\not\equiv 1 \pmod{144}$* — permanently safe by key definition *(including Lk|j3|A8|q73); the unique $q \equiv 1$ class is L3|j4 (its family-level elimination is Proposition 4.14). Every dangerous key $(\cdot, j \geq 1, 2^j \bmod 144, 1)$ contains at most one power, $A = 2^j$ (the key fixes $\nu_2(A) = j$), excluded at the family level: members satisfy $A = 3 \cdot 2^{k-1}w + 2^j\mu \geq 3 \cdot 2^{k-1} \cdot 3 + 2^j > 2^j$ ($w \geq 3$, $\mu \geq 1$).*

Proof. A power of two has odd part 1; (i)–(iii) are immediate, and the 19/20 count is a key-segment check against Proposition 4.11. The magnitude inequality is as displayed. □

Proposition 4.13 (m-record chain and class-generation timetable; declared domain 10^7 anchors)**.** *(Independent full re-run ref-b6, digit-for-digit.) The m-records $m = 1..25$ all occur exactly at $n = 3 \cdot 2^m - 3$ (the pure $w = 3$ doubling chain, direct transfers $n \to 2n+3$: $3 \to 9 \to 21 \to \cdots \to 100663293$), at anchor indices $0, 1, 2, 4, 8, \dots, 8{,}173{,}106$.* ***Of the first*** *25* ***m-records,*** *14* ***create a new class*** *($m \in \{4, 5, 7, 8, 11, 12, 13, 14, 16, 17, 21, 22, 23, 24\}$, plus five additional instants at $w \in \{9, 15, 21\}$); the record anchors $m = 18, 19, 20, 25$ create none. The table is unsaturated through 10^7 anchors (last new class at anchor 4,086,554);* "the class table grows without bound in m" is a trend inference, not a theorem — *the per-m visits of the $w = 3$ chain guarantee unbounded* opportunity *for growth, while per-m generation is content of Assumption M.*

[3] The power-containment decider powInAP of births11 shows 1,722 *false negatives* in a 30,000-case synthetic cross-check (the s=0 branch is skipped entirely for odd D; the discrete-log loop breaks out early) and 249 false positives — harmless for the 14 certified classes (all have $j < s$, the defective branches untriggered; independently re-verified by an enumeration-based decider) but *not to be reused as-is for new classes at $m > 25$*; the correct tool is the preperiod-plus-period enumerator of Lemma 4.10.

4.3. **The drift law M″: the final explicit form of the residual.** Every L-exit root birth satisfies $A = \sigma + r_2/2 + 1 = (3\sigma + d)/2 + 1$ ($d = r_2 - \sigma$). Synthetic experiments (constructed anchors $P(2^a w - 3)$ walked directly in the collapsed engine, $w \equiv 3 \pmod 6$) show that births with $q \equiv 1 \pmod{144}$ occur only in *clean geometric-head windows* — L-configuration run vectors

$$\text{Lk-type } (\sigma = 2^k w{+}1): \quad [1,\ 2^k w{+}c,\ 2^k w,\ 2^{k+1} w,\ 2^{k+2} w{-}1\,],$$
$$\text{L3-relay } (\sigma = 2^k w \text{ exactly}): \quad [1,\ 2^k w{+}c,\ 2^k w{-}1\,]$$
$$\text{or with geometric tail} \quad [1,\ 2^k w{+}c,\ 2^k w,\ 2^{k+1} w{-}1\,],$$

each window k contributing an *affine family*, linear in the window parameter w:

(2)
$$A = 3{\cdot}2^{k-1}\, w{+}2^j \mu, \qquad \mu \in \{1, 145, 289, \dots\} \text{ odd with } \mu \equiv 1 \pmod{144}, \qquad d = 2(m_0{-}1){-}3\varepsilon,$$

with $m_0 = 2^j \mu$ and $\varepsilon \in \{0, 1\}$ the +1-offset of σ (Lk windows $\varepsilon = 1$, L3 windows $\varepsilon = 0$; the values $d = 30, 59, 126, 251, 288, 510, 576, 575$ all match). *The coefficient* 2^{k-1} *is a property of the window* k*, not of the class*: the class L3|j4 in the form $9 \cdot 2^{12} W + 16 = 3 \cdot 2^{12} w + 16$ ($W = w/3$, odd automatically since $w \equiv 3 \bmod 6$) is specific to the $k = 13$ window, and its members occur only in anchor segments with $\nu_2(n + 3) = 13$. Write M′ (the residual arithmetic proposition) for: every L-exit root birth $A = \sigma + r_2/2 + 1$ in every segment of the anchor sequence has odd part $q \geq 3$ (equivalently $A \notin \{2^t : t \geq 2\}$).

Proposition 4.14 (drift law; family census)**.** *(a)* Verified family table *(census:* $a = 13..300 \times w \in \{3, 9, 15\}$*,* 864 *segments, all* 1,683 *births with* $q \equiv 1$ fully classified*, each member verified against the exact identity; synthetic extension* $a = 13..80 \times 6w$*:* 408 *segments,* 7,062 *root births, all* 336 $q \equiv 1$ *births inside* $\{F_{13}, F_{26}, F_{50}\}$*,* 0 *powers):*

window k	*type*	*key*	σ*-form*	d	A*-form*	*first / persist.*	*members*
13	*L3*	*(L3,4,16,1)*	$2^{13}w$	*30*	$3 \cdot 2^{12} w + 2^4$	a=13 *(exact)*	*3*
26	*Lk*	*(Lk,5,32,1)*	$2^{26}w{+}1$	*59*	$3 \cdot 2^{25} w + 2^5$	$a \geq 27$ *(persist.)*	*822*
50	*L3*	*(L3,6,64,1)*	$2^{50}w$	*126*	$3 \cdot 2^{49} w + 2^6$	a=50 *(exact)*	*3*
99	*Lk*	*(Lk,7,128,1)*	$2^{99}w{+}1$	*251*	$3 \cdot 2^{98} w + 2^7$	$a \geq 100$ *(persist.)*	*603*
110	*L3*	*(L3,0,1,1) odd*	$2^{110}w$	*288*	$3 \cdot 2^{109} w + 145$	a=110 *(exact)*	*3*
194	*L3*	*(L3,8,256,1)*	$2^{194}w$	*510*	$3 \cdot 2^{193} w + 2^8$	a=194 *(exact)*	*3*
219	*L3*	*(L3,0,1,1) odd*	$2^{219}w$	*576*	$3 \cdot 2^{218} w + 289$	a=219 *(exact)*	*3*
219	*Lk*	*(Lk,1,2,1)*	$2^{219}w{+}1$	*575*	$3 \cdot 2^{218} w + 2 \cdot 145$	$a \geq 220$ *(persist.)*	*243*

The odd part of each family equals $3{\cdot}2^{k-1-j} w{+}\mu$*, odd: for* k=13, $\geq 768w{+}1 \geq 2305$*; for* $k = 26$, $\geq 3 \cdot 2^{20} w + 1 \geq$ 9,437,185*; the same shape, larger, for* k=99/194*; the odd families (*j=0*) are* $\geq$ 145/289*; the* k=219*-Lk family is* $\geq 3{\cdot}2^{217} w{+}145 \geq 145$ — all eight families are parity-killed *(all* 1,683 $q \equiv 1$ *members pass).* ***(b) Structural facts.*** *(1) The persistent families (*$k = 26, 99, 219$*-Lk) have* σ *and* A *independent of the anchor level* a *(after first appearance every segment hits the same* (σ, A)*, varying only with* w*) — mid-ladder windows (round-law locality: the* α*-walk is autonomous and triggers on passage); the level-exact families (*$k = 13, 50, 110, 194, 219$*-L3) occur only at levels whose endgame lands exactly in that window. (2) The window positions* $\{13, 26, 50, 99, 110, 194, 219\}$ *(the eighth family first appears at level* 220*) form two near-doubling chains (*$13 \to 26 \to 50 \to 99 \to 194$ *and* $110 \to 219$*) — consistent with the timetable of the* $\nu_2(\cdot)$*-record chaos (the non-periodicity clause of Theorem 9.1: new bits injected forever* $\Rightarrow$ *new windows expected forever). (3) Beyond these eight families there is* no other $q \equiv 1 \pmod{144}$ *birth in the census or synthetic range;* 0 *powers; all E1r births divisible by* 3*; all landings* $\equiv 3 \pmod 6$ *and* $\geq n + 6$*. The constructed anchors* $a \leq 80$ *produce several further* $q \not\equiv 1$ *keys (e.g. Lk|j1|A62|q31, first at* a=49*) — all safe by key definition (Theorem 4.12(i)).* ***(c)*** M″ ***(drift law).*** *For every* $m \geq 1$ *and every* $w \equiv 3 \pmod 6$*, the* $q \equiv 1 \pmod{144}$ *L-exit root births in*

the segment of the anchor $n = 2^m w - 3$ belong to finitely many window families $\{F_k\}_{k\in K(m)}$ (K monotone, $K \cap [1, 300] = \{13, 26, 50, 99, 110, 194, 219\}$), each of the form (2) with odd part odd and ≥ 145. ***(d) Equivalence statement.*** *$\mathrm{M}'' \Rightarrow \mathrm{M}'$ ("no L-exit root birth on the orbit has odd part $q = 1$"); conversely, given the enumeration ring (every dangerous key belongs to some window family), $\mathrm{M}' \Leftrightarrow \mathrm{M}''$. Relation to Assumption M: $\mathrm{M} \Rightarrow$ class preservation $\wedge\, \mathrm{M}'$ (Theorem 1.1 + the contrapositive of Theorem 4.8), while $\mathrm{M}' \wedge$ class preservation $\Rightarrow$ non-halting (Corollary 4.16) — so within the established architecture M'' (together with class preservation)* carries exactly the entire non-halting content of Assumption M*: the endgame/round-law part of M determines when windows activate, and M'' compresses M into a single arithmetic sentence in birth coordinates.*

Proposition 4.15 (synthetic assertion table for $m = 26..40$)**.** *Over 60 segments (15 levels $\times\, w \in \{3, 9, 15, 21\}$) of constructed anchors walked directly, all assertions pass: 0 power births (60/60); all E1r divisible by 3; all landings $\equiv 3 \pmod 6$ with $\Delta \geq 6$ (60/60). At $a = 26$:* zero $q \equiv 1$ *members (the k=26 window not yet active; the k=13 key (L3,4,16,1) has no recurrence in 26..40). At $a = 27..40$ (56 $q \equiv 1$ members):* 56/56 satisfy the exact F_{26} identity $A = 3 \cdot 2^{25} w + 32 = 9 \cdot 2^{25} W + 32$ *($W = w/3 \in \{1, 3, 5, 7\}$, all odd — structural from $w \equiv 3 \pmod 6$),* $\sigma = 2^{26} w + 1$, $r_2 - \sigma = 59$, $\nu_2(A) = 5$, *odd part $q = 3 \cdot 2^{20} w + 1$ odd $\geq$ 9,437,185 > 1, hence not a power. Cross-checks: the $a = 13$ segment reproduces the orbital observations term-by-term ($R(24576, 24606) \to A = 36880$; $R(12305, 36886) \to A = 30749$); the orbital members $w = 27, 33, 3501, 6357$ also lie in F_{13} (verified digit-by-digit against $12288w + 16$).*

Corollary 4.16 (conditional assembly; the M'' route)**.** *Assume (i) M' (equivalently M'' under the enumeration ring) and (ii) class preservation (every anchor $n_k \equiv 3 \pmod 6$). Then M, started on the blank tape, never halts.*

Proof. By Theorem 4.8 a halt requires a root birth equal to 2^t. Roots split into: E1-reentries (excluded unconditionally by Lemma 4.9); anchor initializations (n odd $\Rightarrow$ first sweep $y = n - 2$ odd $\Rightarrow$ no birth — guaranteed by class preservation); L-exits (excluded by M'). □

The class-preservation clause is *necessary*: even-anchor births are an independent genuine power channel — at $n = 10, 42, 170, 682, \ldots$ the anchor birth $A = 3n/2+1 = 16, 64, 256, 1024, \ldots$ is exactly a power of two, matching a subfamily of the known even-n halting set $\{10, 26, 42, \ldots\}$.

Remark 4.17 (the three precise gaps; unconditional resolution refused)**.** All assertion tables pass (0 powers, 56/56, all eight families killed, class tables and growth as stated), but the proof chain does not close — exactly three rings are missing: **(1) unbounded window enumeration = M:** the dangerous-key set grows with m (one family $\to$ eight by $a \leq 300$) with no evidence of stopping (the $\nu_2(\cdot)$-record timetable $\Rightarrow$ new windows expected forever); "every family is parity-killed" does not imply "these are all the families" — the $m \leq 2000$ table exhaustion plus synthetic 300 is an empirical closure. **(2) class preservation is conditional** (L7/M-architecture, Corollary 4.16(ii)): even-anchor births are an independent real power channel. **(3) $m > 25$ is a constructed-anchor property:** the synthetic verification uses the collapsed engine (cross-validated by the births10 uncollapsed 0-mismatch check + the independent raw machine on 12/12 anchors exactly), the constructed-anchor segments were not individually re-checked at the raw level, and the "orbital" reading of $m = 26..40$ depends on the orbit actually reaching those anchors (the m-record chain is machine-confirmed through $m \leq 25$; beyond that belongs to M, Proposition 4.13).

Proposition 4.18 (depth-1 birth guards are not closed)**.** *Let I_1 be I_0 together with same-shape algebraic guards at each birth site (odd part of the birth value ≥ 3; the f_2 guard for $G(2, \cdot)$; $w \equiv 2 \pmod 3$ at $A = 3$ sites; exact forms in the* `invariant-casecheck` *suite). Then I_1 contains the*

entire real orbit (trajectory library: 1,357,014 *states, zero violations), but* I_1 *is* not *closed: a closure search over the declared domain (*S/L *tails of length* ≤ 6*) yields* 1,255 *violations, all of a single class — an odd-*r_2 *L arming, via (L1), its S-odd child, whose local guard is blind to the child's next-step birth.*

Proposition 4.19 (per-depth pollution of birth guards; witnessed)**.** *For every tested depth* $d \in \{1, 2, 3\}$ *there is a configuration in* I_0 *whose pipeline has* $d - 1$ *S-even births that are not powers of two, followed by a power-of-two birth, and which truly halts:*

depth	*witness*	*birth sequence*	*phase events*	*raw steps*
d=1	$L[1, 3, 1, 3, 1]$	$8, 4$	*13*	*195*
d=1′	$S[1, 1, 1, 1, 1, 1]$	$8, 4$	*15*	*230*
d=1″	$G[4, 2, 7, 5]$	$16, 8, 4$	*33*	*728*
d=2	$S[3, 2, 25, 43, 23, 12, 4, 2]$	$5, 64$	*152*	*10,313*
d=3	$S[12, 28, 23, 42, 14, 22, 38, 31]$	$27, 71, 128$	*542*	*71,856*

All five witnesses were independently reproduced at the raw-machine level (event-by-event tape agreement with the phase configurations, raw halt confirmation, full chains inside I_0*; additionally, the apparent non-halting witness* $S[39, 34, 5, 30, 43, 19, 37]$ *of the* 400*-step-capped search halts within* 442 *events under a larger cap — a cap artifact, not a counterexample). Moreover* residue-class refinement is impossible*: the bad births* $\{2^t\}$ *have residues mod* 12 *in* $\{4, 8\}$*, the very classes occupied by the orbit's own S-even heads (the* $A \bmod 12$ *histogram covers* $\{0, \ldots, 9\}$*, with* 9,282 *cases* $A \equiv 4$ *and* 396 *cases* $A \equiv 8$*); the intra-class split is exactly* ν_2 *— irreducibly 2-adic. The entry edges of bad births form a* family of birth-entry edges*: (L1) dominant (the* 1,255 *violations of* I_1 *and the* $d = 2$ *witness both enter through it), with (R) also occurring (the* $d = 3$ *witness enters through (R)*$\to S_3$*); the pollution mechanism is the per-level translation of the (S1)/(L1) pipeline (Remark 4.20).*

Remark 4.20 (structural translation for $d \geq 4$; not proved)**.** After each per-level repair (a depth-d guard reads "the first d births of the pipeline are non-powers-of-two"), an edge of the same shape reappears one level deeper in the next L-round: each repair translates the same-shaped pipeline by one level. Per-depth witnesses for every $d \geq 4$ are *expected* from this mechanism (random search stably produces fresh witnesses at each depth), but their per-d existence is precisely the statement at issue and is not claimed here; "no finite guard depth closes" is accordingly stated at witnessed level (Proposition 4.19 plus this mechanism), not as a theorem.

Remark 4.21 (relation to the no-precision conjecture)**.** The companion derivation paper states the no-precision conjecture for coarse templates (per-run residues at a fixed modulus plus linear inequalities); Proposition 4.19 does not prove it and does not change its status. What it supports, at witnessed level (grade B), is the strengthening direction beyond that language: residue classes and algebraic head-form guards alike fail at every finite guard depth. The forcing edges are not on the mod-3 side (Theorem 4.3 seals that side completely) but at the 2-adic birth sites: the entry edges form a family ((L1) dominant, (R) also occurring), with the (S1)/(L1) pipeline translating one level deeper per repair (Remark 4.20).

5. Landmarks and classes

Fix an odd *unit* $w \geq 3$ for the rest of the paper.

Definition 5.1 (landmark)**.** For $\alpha \equiv 1 \pmod 6$, $\alpha \geq 7$, $t \geq 9$, and slack $s \geq 0$, let $E := 2^{t-3}w$ and

$$L(\alpha, t; s) \;=\; G\big(\alpha + 3 \cdot 2^{t-3}w,\; \alpha - 1 + 5 \cdot 2^{t-3}w,\; 2^t w,\; 2^{t+1}w,\; \ldots,\; 2^{t+s}w,\; 2^{t+s+1}w - 1\big).$$

Its *class* is $h = \nu_2(\alpha + 1)$.

Lemma 5.2 (class parametrization)**.** *Fix $h \geq 1$. The integers $\alpha \equiv 1 \pmod 6$, $\alpha \geq 7$ with $\nu_2(\alpha + 1) = h$ are exactly*

$$\alpha \;=\; 2^{h+1}(3c + \rho_h) + 2^h - 1, \qquad c \geq 0,$$

where $\rho_h \in \{0, 1, 2\}$ is the unique residue with $(-1)^{h+1}\rho_h + (-1)^h \equiv 1 \pmod 3$.

Proof. $\alpha + 1 = 2^h\,\omega$ with ω odd forces $\omega = 2(3c + \rho) + 1$ with $\rho \in \{0, 1, 2\}$, $c \geq 0$, giving the shape. Modulo 3, using $2 \equiv -1$: the displayed congruence is the condition $\alpha \equiv 1$, which singles out one ρ since the three values of ρ give the three residues. □

Theorem 5.3 (round law)**.** *There exist functions $\Delta\alpha, \Delta t : \mathbb{N}_{\geq 1} \to \mathbb{N}$ with $\Delta\alpha(h) \equiv 0 \pmod 6$ and $\Delta t(h) \in [2, 9]$ such that for every landmark $L(\alpha, t; s)$ of class h:*

1. *if $s \geq \Delta t(h)$, the phase trajectory returns to the landmark $L\big(\alpha + \Delta\alpha(h),\, t + \Delta t(h);\, s - \Delta t(h)\big)$, and no halting occurs;*
2. *if $s < \Delta t(h)$, the trajectory exits to the anchor $P(n')$ with $n' = 2^{t+s+1}w + c$, $c \equiv 3 \pmod 6$, $c \geq 3$.*

The proof runs through §5.1–5.3 by strong induction on h; it is completed in §5.3.

5.1. **The halving cascade.**

Lemma 5.4 (cascade)**.** *Let $L(\alpha, t; s)$ have class h, write $\alpha + 1 = 2^h\omega$ (ω odd), assume $t \geq h + 4$, and put*

$$\Psi \;=\; \omega + 3 \cdot 2^{t-3-h}w.$$

Then Ψ is odd, and the first h events of the round are the halving steps with leading heads

$$a_i \;=\; 2^{h-i}\,\Psi - 1, \qquad i = 0, 1, \ldots, h,$$

so that the h-th of them lands on the configuration

$$S_0 \;=\; G\big(\Psi - 1,\ \Psi - 1,\ 2\Psi,\ 4\Psi,\ \ldots,\ 2^{h-1}\Psi,\ \underbrace{\alpha + 5E}_{\beta},\ 2^t w,\ \ldots,\ 2^{t+s+1}w - 1\big),$$

i.e. on $\Phi(\Psi, h)$ with its final run $2^h\Psi - 1$ replaced by the triple (β, E-geometric tail, -1 terminator), $\beta = \alpha + 5E$.

Proof. The round opens with rule (G≥) on the leading head $a_0 = \alpha + 3 \cdot 2^{t-3}w$. Since $\alpha \equiv 1 \pmod 6$ and $3 \cdot 2^{t-3}w \equiv 0 \pmod 6$ ($t \geq 9$), a_0 is odd, and by the halving arithmetic of Theorem 3.4 the successive leading heads are $a_i = (a_0 - \delta_i)/2^i$-type where the affine bookkeeping gives $a_i + 1 = 2^{h-i}\Psi$: indeed $a_0 + 1 = \alpha + 1 + 3 \cdot 2^{t-3}w = 2^h\omega + 3 \cdot 2^{t-3}w = 2^h\Psi$ by definition of Ψ, and the halving map divides $a + 1$ by 2 at each step while leaving the tail invariant, provided each intermediate head is odd — which holds because $a_i + 1 = 2^{h-i}\Psi$ with Ψ odd. Meanwhile the displaced material accumulates as the geometric tail $(2\Psi, 4\Psi, \ldots)$ exactly as in the prologue. At $i = h$ the head is $\Psi - 1$ and the configuration is S_0 as stated; the run $\beta = \alpha + 5E$ arises because the second leading run of L is $\alpha - 1 + 5E$ and the cascade transfers the increment $+2$ net. (Each such identity was verified symbolically and on $h = 1, \ldots, 60$ by exhaustive run-by-run comparison.) □

Lemma 5.5 (entry to ladder scale)**.** *For $h \geq 12$, the trajectory from S_0 reaches, after reading only runs up to exponent $\leq 10 < h - 1$, the landmark-scale configuration $L(7, 9)$-type at unit Ψ with the modified tail $(\beta, E\text{-tail}, -1)$ carried verbatim.*

Proof. From S_0 the machine consumes the Ψ-geometric runs in the fixed manner of a fresh $\Phi(\Psi, h)$-trajectory; the first landmark of any Φ-trajectory is $L(7, 9)$-shaped (leading counter 7, time parameter 9), reached having read only the runs $2\Psi, \ldots, 2^9\Psi$. The modified tail sits entirely at exponents $\geq t > h + 4 \geq 16 > 10$, hence beyond the read window and carried verbatim. (Symbolic derivation; witnessed on $h = 12, \ldots, 60$.) □

5.2. **The universal ladder.**

Definition 5.6 (ladder)**.** Define the ladder as the sequence $(A_j, \tau_j)_{j\geq 0}$ with $(A_0, \tau_0) = (7, 9)$ and

$$(A_{j+1}, \tau_{j+1}) \;=\; \big(A_j + \Delta\alpha(H_j),\; \tau_j + \Delta t(H_j)\big), \qquad H_j := \nu_2(A_j + 1),$$

where $\Delta\alpha, \Delta t$ are the functions of Theorem 5.3.

Lemma 5.7 (autonomy)**.** *On any Φ-trajectory (with any admissible modified tail), the successive landmarks follow the ladder: the j-th landmark is $L(A_j, \tau_j; \cdot)$ at the unit scale, with the tail parameters evolving independently. Consequently the sequence of classes visited depends only on j, not on the unit or the tail.*

Proof. By Theorem 5.3 the passage landmark→landmark changes (α, t) by $(\Delta\alpha(h), \Delta t(h))$ with h the class of the current α; the unit and tail enter only through the slack bookkeeping (which decides landmark-vs-exit, not the map itself). □

Lemma 5.8 (margins and class bound)**.** *The ladder satisfies, for all computed positions:*

(1) $\tau_j \;\geq\; 2H_j + 16$ *for all* $j \geq 3$*;*
(2) $A_j \;\leq\; \tau_j^2$ *for all* $j \geq 8$.

Moreover, if these inequalities hold up to position j, they persist: $A_{j+1} \leq A_j + 2H_j + 6 \leq \tau_j^2 + 4\log_2 \tau_j + 6 \leq (\tau_j + 2)^2 \leq \tau_{j+1}^2$ using $\Delta t \geq 2$.

Proof. Induction as displayed, given the band bound $\Delta\alpha(H) \leq 2H+6$, which holds on all classes where the round law has been established (this is part of the induction of §5.3: the induction hypothesis at class h includes the band bound for all classes $\leq h$). Items (1)–(2) were verified exhaustively on the first 200,000 ladder positions, which cover all $\tau_j \leq 10^6$. □

Corollary 5.9 (well-foundedness)**.** *For $h \geq 16$, every ladder position with $\tau_j \leq h - 1$ satisfies $H_j \leq 2\log_2 h < h$: the ladder prefix traversed inside a round of class h involves only rounds of class strictly smaller than h.*

5.3. **The master segment.** Fix $h \geq 16$ and a landmark $L(\alpha, t; s)$ of class h. By Lemmas 5.4–5.5 the round consists of: the cascade into S_0; a ladder prefix from $L(7, 9)$-scale up to the position j_0 = the last j with $\tau_j \leq h-65$ (all of whose rounds have class $< h$ and are covered by the induction hypothesis); and the *master segment*: the events from the ladder literal $L(A_{j_0}, \tau_{j_0})$-scale to the end of the round.

Computational Assumption 5.10 (master closure)**.** For every $h \geq 1$, the symbolic derivation from the ladder literal $(A_{j_0(h)}, \tau_{j_0(h)}, h)$ with the outer parametrization of Lemma 5.2 ($c \geq 0$, $u \geq 0$, $w = 6u + 3$) and $t \geq 2h + 16$ is *single-path*: every branch decision (parity and run-size comparisons) is forced uniformly over the entire domain, no split ever occurs, and the derivation terminates in $O(\mathrm{poly}(\log h))$ events at either the next real landmark (yielding $\Delta\alpha(h) \equiv 0 \bmod 6$, $\Delta t(h) \in [2, 9]$) or, if $s < \Delta t(h)$, at the anchor exit of Theorem 5.3(ii). Moreover the derivation is *h-uniform*: its complete form set depends on h only through the ladder literals (A_{j_0}, τ_{j_0}) and the exponent offsets of the modified tail (the uniformity clause, spelled out in Remark 5.11); this is what allows the finite audits below to be lifted from the verified classes to all h.

Remark 5.11 (the uniformity clause and its evidence)**.** The h-uniformity clause of Assumption 5.10 states that the master derivation depends on h only through the ladder literals (A_{j_0}, τ_{j_0}) and the exponent offsets of the modified tail, whose margins grow monotonically with h (the read window is bounded by $\Delta t + 3$ while the tail begins at exponent $t \geq 2h + 16$). Under this clause, any failure of the single-path property at some $h_* > 10^6$ would be witnessed by a violation of one of finitely many margin inequalities, each monotone in h beyond the verified range; the clause was verified concretely on all classes $h \leq 2000$ (exhaustively) and on sampled classes through $1.2 \cdot 10^7$. It is part of the assumption, not a separate theorem: the monotonicity argument above is a structural observation over the verified derivation shapes, and the safety proof cites it as such (conditional on Assumption 5.10).

Proposition 5.12 (evidence for Assumption 5.10)**.** *Assumption 5.10 (single-path clause and uniformity clause alike) was verified by exhaustive single-path symbolic derivation: (i) for all* $h = 16, \ldots, 1024$*, by direct agreement with the independent per-h direct engine (*1009/1009*); (ii) for* $h = 1025, \ldots, 2000$ *exhaustively and for sampled h up to* $1.2 \cdot 10^7$ *(*976 *table rows plus on-the-fly ladder extensions, all single-path, all law-shape consistent); (iii) at eight classes* $h \in [1028, 1777]$ *against numeric ground truth (exact; the class* $h = 1538$ *among them realizes the rare value* $\Delta t = 9$*, correcting the earlier bound* $[2, 8]$ *— see the erratum in the dossier); (iv) at fresh classes* $h \in \{1823, 2048, 31337, 666667\}$ *— at* $h = 1823$ *with three-way agreement (master = direct engine = precomputed table) — and at* $h = 45678$ *via the published blind-test recipe, and — with the ladder recurrence computed on the fly rather than stored, cross-checked against the stored ladder at* $h = 1777$ *— at the fresh classes* $h \in \{9999991, 10^7, 10^7{+}3, 12345679\}$*, each single-path with* $\Delta\alpha/h \approx 1.3382$*.*

Proof of Theorem 5.3. By strong induction on h. *Base* $h \leq 15$*:* direct single-path symbolic derivations (the first 15 rows of the per-h table). *Step* $h \geq 16$*:* the decomposition of §5.1–5.3 expresses the round as cascade (Lemma 5.4, algebraic) + entry (Lemma 5.5) + ladder prefix (rounds of class $< h$, each correct by the induction hypothesis and Corollary 5.9) + master segment (Assumption 5.10). The composite is single-path and ends in outcomes (i) or (ii). The band bound $\Delta\alpha(h) \leq 2h{+}6$ required by Lemma 5.8 is carried along the same induction: it holds on the base, and the master outputs satisfy it on all verified classes ($\Delta\alpha/h \approx 1.338$ throughout, e.g. $\Delta\alpha(10^6) = 1338210$). □

6. Endgames

Theorem 6.1 (anchor exit)**.** *Let* $L(\alpha, t; s)$ *of class h occur on the* $\Phi(w, m)$*-trajectory started from the anchor* $P(n)$ *(*$n + 3 = 2^m w$*), and let* $s < \Delta t(h)$*. Then s is the slack to the* terminator *run* $2^m w - 1$ *of* Φ*, i.e.* $t + s + 1 = m$*, and the trajectory exits the* Φ*-family at the anchor*

$$n' \;=\; 2^m\, w + c \;=\; (n+3) + c, \qquad c \equiv 3 \pmod 6,\ c \geq 3,$$

so that $n' \geq n + 6$*.*

Proof. Along a Φ-trajectory the slack s counts the geometric tail runs beyond the current read window; the trajectory ends in an exit exactly when the read window of the current round overruns the terminator $2^{t+s+1}w{-}1$, which is the final run $2^m w{-}1$ of $\Phi(w, m)$; hence $t{+}s{+}1 = m$. The sweep arithmetic then resolves the configuration into a two-run $R(n', n')$ with $n' = 2^m w{+}c$. The congruence $c \equiv 3 \pmod 6$ and $c \geq 3$ were verified symbolically for every endgame cell (all $h \leq 160$ exhaustively — 841/841 cells, including two independent re-audits; the full $h \leq 1024$ grid adds 5049 further cells, all passing, for the ledger total 5890), with the endgame anchor form additionally matched against constructed landmark walks (216/216). The growth inequality follows: $n' - (n + 3) = c \geq 3$, i.e. $n' \geq n + 6$. □

7. No halting

Theorem 7.1 (safety)**.** *No configuration on any trajectory from $P(3)$ is of the form $G(1,\cdot)$ or $G(3,\cdot)$; in particular no G-state has leading head $a \in \{1,3\}$, and by Lemma 2.4 M never enters the halting transition. (The master-segment part of the computational audit below is conditional on Assumption 5.10 in the standard way.)*

Lemma 7.2 (exhaustiveness of the state classification)**.** *Every G-state occurring on a trajectory from $P(3)$ occurs inside one of the segment derivations of Theorem 3.4, Lemmas 5.4–5.5, the rounds of Theorem 5.3 (base table for $h \le 15$; ladder prefix rounds and master segment for $h \ge 16$), or the endgame cells of Theorem 6.1; and each such segment derivation is single-path, hence enumerates its events exactly.*

Proof. By the assembly recursion of §8, the trajectory from $P(3)$ is the concatenation of the segment derivations of the anchor transitions. Each segment is single-path: the prologue and cascade are deterministic algebra (Theorem 3.4, Lemma 5.4); the base rounds $h \le 15$, the ladder-prefix rounds (classes $< h$, induction), the master segments (Assumption 5.10), and the endgame cells are single-path symbolic derivations by construction/verification. A single-path derivation from a fixed initial configuration enumerates exactly the events that occur. □

Proof of Theorem 7.1. We split the argument into an algebraic part and a computational part.

Algebraic part (prologue, entry, and cascade families). Let a be the leading head of a G-state in: (a) the prologue halving chain, where $a_i + 1 = 2^{m-i}w$ with $w \ge 3$ odd (note $w \ge 3$ since $n \equiv 3 \pmod 6$ forces $n + 3 \ge 6$); the prologue's initial head $n + 2 \ge 5$ is odd and $\ne 3$ as $n \ge 3$; (b) the halving cascade of a round with class h and $t \ge h + 4$, where $a_i + 1 = 2^{h-i}\Psi$ with $\Psi = \omega + 3 \cdot 2^{t-3-h}w$; here $\omega \ge 1$ is odd and $3 \cdot 2^{t-3-h}w \ge 3 \cdot 2 \cdot 3 = 18$, so $\Psi \ge 19$ is odd; (c) a landmark G-head $\alpha + 3 \cdot 2^{t-3}w$ with $\alpha \equiv 1 \pmod 6$, $\alpha \ge 7$, and $3 \cdot 2^{t-3}w \ge 3 \cdot 64 \cdot 3 > 12$: the head is odd (odd + even) and ≥ 19. In cases (a)–(c), if a is odd then $a + 1 = 2^j \cdot (\text{odd} \ge 3)$ for some $j \ge 1$, while $a \in \{1,3\}$ would require $a + 1 \in \{2,4\} = 2^j \cdot 1$ with odd part $1 < 3$: contradiction. The even values that occur ($a = w - 1$ with $w \ge 3$ odd) are ≥ 2 and even, hence also not in $\{1,3\}$. So $a \notin \{1,3\}$ throughout these families.

Computational part (intermediate and master/endgame states). The G-states inside sweeps, master segments, and endgames are exactly those enumerated by the single-path derivations of Lemma 7.2. The 2-adic audit (exact two-variable (u,c) solvers) checked, over 273,654 symbolic forms — all round classes $h \le 1024$, all endgame cells, and the full master-path form set (which, by the uniformity clause of Assumption 5.10 (Remark 5.11), is h-uniform) — that no form has a leading head with $a + 1$ a power of 2 (this excludes $a \in \{1,3,7,15,\dots\}$ simultaneously, in particular both halting heads) or reaches such a head under any admissible continuation of its symbolic domain: 0 dangers, 0 undecided.

By Lemma 7.2 the two parts cover every G-state on the trajectory, and in neither part does a head in $\{1,3\}$ occur. By Lemma 2.4, M never enters the halting transition. □

8. Assembly and proof of the main theorem

Proof of Theorem 1.1. By Lemma 3.2 the run reaches $P(3)$ at step 54. Define the anchor sequence $n_0 = 3$ and, inductively, $n_{k+1} := n'$ the exit anchor of Theorem 6.1 applied to the $\Phi(w_k, m_k)$-trajectory started from $P(n_k)$ (which exists by Theorem 3.4 with $n_k + 3 = 2^{m_k}w_k$). We maintain the invariant $6 \mid n_k + 3$. It holds for $n_0 = 3$. Suppose $6 \mid n_k + 3 = 2^{m_k}w_k$; then $3 \mid w_k$, and $6 \mid 2^{m_k}w_k$ (both 2 and 3 divide it). The exit of Theorem 6.1 gives

$$n_{k+1} + 3 \;=\; 2^{m_k}w_k + (c_k + 3), \qquad 6 \mid c_k + 3 \;\; (\text{since } c_k \equiv 3 \pmod 6),$$

so $6 \mid n_{k+1}+3$. In particular every $n_k \equiv 3 \pmod 6$ (odd), which is the domain of Theorem 3.4. Theorem 5.3 gives the complete decomposition of every transition into prologue + rounds + exit, with $n_{k+1} \geq n_k + 6$ (Theorem 6.1); the sequence is defined for all k; and by Theorem 7.1 no halting configuration ever occurs. Since the phase calculus of Lemma 2.3 mirrors the machine exactly (including the halting transition), M never halts. □

9. Computational components

The following components are carried by exhaustive symbolic computation and are stated precisely so that they can be re-audited or formalized independently (scripts in the repository; each row is a one-line rerun):

Component	Domain	Result
Rule-level grid (Lemma 2.3)	$a \leq 20 \times b \leq 10\times$ tails	564/564 (`rulecheck`)
Trajectory faithfulness (Lemma 2.3)	956 anchors (n=2..957)	955 equal; $R(2,2)$ off-grammar
Prologue (Theorem 3.4)	$n \leq 6003$; $m \leq 1200$	1052/1052
Cascade/entry (Lemmas 5.4,5.5)	$h \leq 60$	327/327
Ladder margins (Lemma 5.8)	$j < 200000$	199997+199992
Base round law	$h \leq 15$	single-path
Induction step	$h = 16..1024$	1009/1009
Master extension	$h = 1025..1.2 \cdot 10^7$	976 rows + 4 fresh
Endgame cells (Theorem 6.1)	$h \leq 160$ (≤ 1024)	841/841 (5890 total)
Endgame forms vs. walks	216 cells	216/216
2-adic safety audit	273654 forms	0 dangers
Ledger closure (Theorem 4.3)	boundary + 2^{40} random	219,398 author + 20,508 independent steps ($k \leq 10$), 0 violations
Ledger orbit (Theorems 4.3, 4.6)	10^7 events	0 violations; 2,908 births, 0 powers
Independent raw re-check	$5 \cdot 10^5$ events	647 births, 0 powers
Root partition (Theorem 4.8; ref-b1)	2×10^7 uncollapsed events, 572 anchors	0 provenance/inheritance/odd-part violations; raw 12/12 anchors
E1-reentry scan (Lemma 4.9; ref-b2)	10^6 anchors	2,787,614/2,787,614 with $3 \mid A$; 3,669,365 re-entries all $b \equiv 2 \bmod 3$
Birth census (Proposition 4.11; births5–7)	10^7 anchors	30,791,021 roots, 0 powers (L-exit direct; E1r by lemma + 10^6 scan)
Uncollapsed cross-check (births10)	600 anchors, 9.5×10^6 events	4,322 births, 0 mismatches
Class certification (Proposition 4.11; births11, ref-b3/b4)	20 classes	14 AP-certified + 3 residue supersets; Lk|j3 single point; L3|j4 family ($W \leq 2 \times 10^6$, 0 powers); pairing 14,560/14,560
m-records (Proposition 4.13; ref-b6)	10^7 anchors	$m = 1..25$ all at $3 \cdot 2^m - 3$; 14/25 create classes
Family census (Proposition 4.14; v7-census)	$a = 13..300 \times \{3, 9, 15\}$	864 walks; 1,683 $q \equiv 1$ all in 8 families; parity kill all
Synthetic families (v7-synth)	$a = 13..80 \times 6w$	408 walks; 7,062 births; 336 $q \equiv 1$ in $\{F_{13}, F_{26}, F_{50}\}$; 0 powers
m=26..40 assertions (Proposition 4.15)	15 levels $\times 4w$	60 segments: 0 powers; 56/56 exact F_{26} form
Depth-1 guard search (Proposition 4.18)	S/L tails ≤ 6	1255 violations, one class
Cascade witnesses (Proposition 4.19)	5 chains	13/15/33/152/542 events, raw halt

The residual in its final form. Magnitude decisions are settled unconditionally by dominance ($h \geq 82$; $h \leq 81$ pointwise); $\Delta t(h) \in [2, 9]$ on all verified classes (Theorem 5.3); the halting branch is unreachable and perturbation-robust (2980/2980 single/sparse-flip counterfactuals close safely; adversarial profiles diverge but are 2-adically unrealizable and never halt). What remains is Assumption 5.10 beyond the exhaustive range ($h \leq 2000$ exhaustive, sampled to $1.2 \cdot 10^7$). A finite-state reduction (`sigdyn.mjs`, `brent32.mjs`) shows this residual is exactly the continuation of the *record timetable* of the autonomous walk: first arrivals of new class depths occur at rounds climbing by a factor ≈ 4 per level (measured schedule 17@19,024, 19@76,096, 21@304,383, 22@1,217,531, 24@4,870,122, and depth ≥ 32 by step 1,246,751,041 of the residue kernel, which terminates at a bad residue — the record-32 event — with no cycle). In birth coordinates the same residual takes the explicit form M″ (Proposition 4.14): the dangerous-key families grow with the window index exactly as the record timetable prescribes, and "never halts $\Leftarrow$ M″ + class preservation" (Corollary 4.16) matches the master-closure chain in non-halting content (Remark 4.17 for the precise three-ring difference).

Theorem 9.1 (periodicity excluded on the verified slice; conjecture beyond)**.** *On the* 32*-bit slice of the signature dynamics (the residual kernel of the* K*-slice model at* $K = 32$*): Brent cycle detection over* $1.25 \cdot 10^9$ *steps finds no cycle, and the kernel terminates at a bad residue (the record-*32 *event); the ladder residues over* $2.4 \cdot 10^6$ *positions are pairwise distinct. Within this architecture, any* K*-bit cycle of the kernel would suppress the record timetable below depth* K *forever, contradicting the observed climb on the verified slice. Consequently no finite closure of Assumption 5.10 exists at* $K \leq 32$*; the corresponding statement for* $K > 32$ *— equivalently, that the full signature dynamics is not eventually periodic — is a* conjecture *(the record climb is measured, not proved, beyond the slice). The residual is equivalent to the* ν_2*-chaos of the autonomous walk — the irreducible core the architecture avoids needing everywhere except at record events.*

Formalizing Lemma 2.3 and the dominance inequality in a proof assistant would raise the conditional theorem and the envelope to proof-assistant grade; eliminating Assumption 5.10 itself requires, by Theorem 9.1, mathematics beyond this architecture (or unbounded per-class computation).